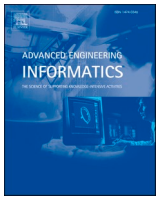
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Automated "E"-aware data processing for construction ESG using building information modeling and large language model

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ABSTRACT

Environmental, Social and Governance (ESG) assessment and disclosure are critical for architecture, engineering, and construction (AEC) companies to market their financial results, reputational position, and compliance with regulatory requirements. Within this framework, the environmental ("E") dimension presents unique and formidable data management challenges distinct from social and governance aspects. Specifically, the complex interplay of quantitative metrics and qualitative descriptions within 'E'-aware data (e.g., measurable resource consumption alongside descriptive material sourcing practices, emissions figures coupled with compliance narratives), amplified by its sheer volume and the persistent ambiguity of environmental indicators and reporting standards, poses significant obstacles to effective 'E'-aware data disclosure. Large Language Models (LLMs) possess inherent advantages in processing such complex environmental information due to their proficient language processing and generalization capabilities. Nonetheless, the development of LLM-based methods explicitly tailored for environmental data management within the construction sector remains underexplored. To this end, this study introduces an automated, LLM-enhanced "E"-aware data processing approach for the construction industry. The innovation of this framework is threefold. First, fifteen "E"-aware indicators are meticulously crafted to align with the specific needs of construction entities. Second, an "E"-aware algorithm, integrated within the Building Information Modeling (BIM) framework, is devised to streamline the aggregation and quantification of environmental data. Third, an LLM-enhanced complex structured data processing mechanism using retrieval augmented generation (RAG) is proposed to facilitate the efficient processing of "E"-aware data pertinent to construction projects. An illustrative case study is employed to validate the feasibility and efficacy of the proposed methodology. The results demonstrate that the developed automated RAG-LLM enhanced framework significantly advances current practice by: (1) enabling standardized "E"-aware data specifications and source mapping; (2) drastically reducing processing time for large-scale ESG documentation (saving 64.4% of time); and (3) providing a robust solution for handling multi-source, multi-format data, thereby enhancing the efficiency and reliability of environmental management and ESG disclosure in the AEC industry.

1. Introduction

The significance of Environmental, Social, and Governance (ESG) assessment in construction projects is escalating within the Architecture, Engineering, and Construction (AEC) industry. The outcomes of ESG assessments are increasingly influencing corporate investments and stakeholder interests [1,2]. Effective ESG management enables companies to fulfill regulatory requirements and this facilitates financial and reputational advancements [3]. Corporate ESG disclosures contain a large amount of environmental information, which is consistent with the increasingly stringent environmental protection requirements for construction projects. Environmental information in corporate operations

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affects the company's long-term financial status and poses risks to survival [4]. Existing research on environmental management and assessment of construction projects mainly focuses on building pollution management [5], low-carbon design [6], human-related comfort evaluation [7] and etc. These studies have helped people improve construction processes and building management methods, but they cannot provide comprehensive information support for complex ESG assessments. For the environmental performance of ESG in construction projects, in addition to data-based construction activity statistics, the biodiversity of the construction site, climate policy and the recycling strategy of the company are also important components of the assessment [8,9]. However, the collection and integration of information in this regard is still imperfect [10]. Although practitioners and researchers have noted the importance of environmental factors in ESG to construction companies, there are still considerable challenges in integrating ESG considerations into management systems in the construction sector.

The main reason that hinders the implementation and research of ESG management in the construction industry is that ESG data itself has its own particularity. The main characteristics of environmental, social and corporate governance information are complex data sources, diverse data formats and difficulty in collection [11]. Compared with the subjective evaluation of S/G based on corporate behavior and the evaluation model driven by humanities [9], the environmental data ("E"-aware data) of construction projects comes from construction materials and energy flows, etc., which are measurable [12]. At the same time, the environmental impact of construction projects is continuous, which is significantly different from the governance indicators based on quarters or audit years [13]. This study focuses on the former because it has a computable basis and has native support advantages in digital construction management [14]. However, for the various ESG disclosure frameworks on the market, the environmental indicators that need to be included in ESG management are not the same [15]. The ambiguity of indicators and data sources of these different frameworks constitutes a major obstacle. Existing ESG assessment and management systems usually adopt a common design across industries for environmental indicators, while the environmental impact of construction activities in construction projects is difficult to accurately collect and manage. This brings difficulties to the management of ESG-related environmental data in construction projects. Meanwhile, the intricate nature of information pertinent to construction projects, coupled with voluminous datasets, necessitates precise collection methodologies to ensure the accuracy of a firm's ESG assessment outcomes [11]. Furthermore, the extensive processing required to render complex data applicable for ESG purposes hedges in elevated labor expenditures and operational inefficiencies. Comprehensive ESG management mandates the incorporation of diverse datasets, including carbon emissions, supply chain logistics, and green technology implementations [2]. The disparate systems and formats from which these datasets originate complicate their direct provision to ESG management and assessment personnel, thereby exacerbating the intricacies of environmental evaluations within the industry [16].

Existing digital solutions in construction, such as Building Information Modeling (BIM), provide support for ESG environmental information management. BIM's primary advantage lies in its ability to streamline information management by integrating teams, workflows, and data across the entire project lifecycle—from conceptual design and engineering to construction and operational phases [17]. This integration positions BIM as an effective management model for processing "E"-aware data. As an IT solution, BIM extended its utility to ESG-related management. In the context of construction projects, BIM data undergirds the preemptive ESG assessment of green buildings in the design phase [18]. Similarly, in the domain of prefabricated construction, BIM aided in analyzing ESG impacts [19]. BIM provides a powerful framework for data management in the AEC industry, which shows its potential when dealing with structured "E"-aware data. However, it is inadequate when dealing with unstructured, context-related documents, which complicate pre-data processing. As an important part of ESG assessment, unstructured data such as policies and reports also need to be processed in a refined manner. Large language models (LLMs) emerge as a promising solution to manage environmentally pertinent content effectively. LLM has already shown its prowess in facilitating cross-departmental communication and enhancing work efficiency within the AEC sector, evidenced by its capabilities in complex information processing [20]. In practical applications, LLM aids in areas such as knowledge mining of construction-related content [21], information retrieval [22] and construction of virtual assistants [23]. As a branch of natural language processing (NLP) technology, LLMs has played an important role in promoting unstructured text processing in the AEC industry. Despite these advances, the application of LLM in ESG-related fields faces challenges due to complex data requirements and imprecise indicator design in ESG assessment. First, there is a lack of domain knowledge for "E"-aware data management of existing construction projects. This difficulty is caused by unclear ESG assessment indicators and unclear related data sources. Second, there is a lack of automated LLM-base0d "E"-aware data management processes. Existing data collection and management solutions include those based on the Internet of Things [24], blockchain [25], etc., but these have limited effectiveness in processing unstructured "E"-aware data such as reporting and corporate strategies. At present, LLM has demonstrated its effectiveness in processing structured ESG data, but this mainly comes from disclosed ESG reports, which have been formatted according to the disclosure framework [26]. NLP solutions for managing construction information are limited to certain specific performance, which are not sufficient for comprehensive information management and have limited effectiveness for construction content with complex data structures [11,27]. Thus, the current research system mainly has two limitations: (1) Lack of domain knowledge on the sources of "E"-aware data in AEC, and (2) lack of automatic "E" data extraction approach for ESG in AEC.

This study aims to propose an automatic "E" data extraction approach in construction for efficient collection and processing of ESG in construction. This approach will explore the information required for ESG environmental assessments and LLM enhanced "E"-aware building data processing solutions. Three objectives will be accomplished:

- **To identify the ontology for the source of "E"-aware data in AEC.** As a basic step in developing the framework, the requirements and definitions of "E" data are designed based on commonly used ESG assessment indicators on the market. This can help managers of construction projects clarify the sources of key information and its relationship to ESG.
- **To develop a BIM-based "E"-aware data extraction algorithm.** This step aims to obtain formatted and digitized ESG-related content after confirming the information source, including accurate carbon emission information and pollution coefficients. An algorithm for accurate "E"-aware data quantification of BIM models based on visual programming will be developed.
- **To propose an LLM-enhanced data processing mechanism for complex structured data using RAG.** This step aims to efficiently process complex data that combines structured data with unstructured data. Based on the developed conceptual framework, further development is made for "E"-aware data in construction and the formation of building ESG reports. It also includes designing a management model for construction project data flow and developing an LLM-enhanced "E"-aware data processing mechanism for the AEC industry.

The remainder of this paper is organized as follows. Section 2 introduces the technical background and highlights research gap. Section 3 presents details of the "E"-aware data automatic management framework for construction and development. Section 4 illustrates framework feasibility and evaluates framework performance. Section 5 discusses the novelty, potential impacts, and limitations of the carbon emission

quantification and dynamic analysis framework. Finally, conclusions and future work are presented in Section 6. Notably, the approach is designed as a generic prototype that also applies to other projects in other countries and regions. It incorporates core features of BIM design and refers to common ESG indicator systems in the global market and meets some standards (such as ISO 19650, ISO14000, etc.).

2. Literature review

2.1. ESG in AEC industry

The concept of ESG first emerged from a report by United Nations Global Compact in 2004 [28], marking the inaugural mention of the term 'ESG'. The contemporary ESG framework encompasses disclosure, evaluation, and investment [29]. Companies adhere to established disclosure standards, providing pertinent information. Subsequently, rating agencies assess this disclosed data, enabling investors to gauge corporate risks and inform their investment decisions, thereby establishing the operational mechanism of ESG. Compared with traditional environmental and sustainability assessments, ESG's environmental performance is often related to a company's operations and development strategy [30]. A significant area of interest is the correlation between ESG metrics and corporate securities prices [31]. This line of inquiry explores how ESG indicators can affect investor perceptions and, consequently, the market value of corporate securities. At the same time, different ESG disclosure frameworks have different requirements for corporate information [32]. Complex indicators and related information make it possible to evaluate the relationship between corporate behavior and ecological and economic development [33]. Within the realm of corporate sustainability reporting, environmental indicators in ESG show more impact on financial and ultimately ESG performance [16]. ESG assessment and disclosure can provide a basis for corporate sustainable development.

Despite these studies, current research on the processing and management of ESG-related data is still insufficient in the construction field. "The Global Status Report for Buildings and Construction" published by the UN Environment Programme [34], indicates that the construction sector was responsible for 37 % of global carbon dioxide emissions in 2022. This statistic has spurred increased market interest in green buildings, prompting nations and regions worldwide to implement policies that incentivize sustainable construction practices and contribute to energy conservation and emissions reduction [35], which also promotes ESG management, investment, and practices within construction firms [36]. Existing work cannot achieve comprehensive and rapid processing of complex data in AEC projects. In the long run, the accumulation of environmental factors such as carbon emissions will further affect the ESG performance of companies [37]. In the AEC industry, construction projects account for the main environmental impact matters of enterprises, but it is difficult to manage them in an orderly manner under the existing ESG assessment framework [11,38]. The crux of the issue lies in the generic nature of most ESG frameworks, which are tailored to suit broad industry standards rather than the specificities of some industry [39]. Consequently, certain indicators lack the granularity necessary for accurate representation, while others may not be entirely applicable to the construction context. Moreover, the intricate web of data sources and formats inherent to construction projects further complicates the acquisition of pertinent "E"-aware data [40]. In summary, there are currently two main problems with "E"-aware data management in the AEC industry: the indicators are unclear, and the relevant data formats are complex and difficult to process.

2.2. "E"-aware data management in AEC

In the environment-related management of enterprises, a large amount of research focuses on environmental protection policies and corporate strategies. For example, the exploration of the correlation between corporate external factors and internal governance behaviors on corporate environmental pollution [41], and the evaluation scheme of corporate environmental performance [42,43]. In the green transformation of construction companies, research on factors affecting corporate environmental performance [44]. Among the existing environmental management programs in the AEC industry, some studies have also guided the development direction of the industry. For example, it analyzes the role and impact of the construction and building sustainability assessment of the United Nations' Sustainable Development Goals (SDGs) [45]. At the same time, some technical means have also been implemented in related management, such as robotics [46] and machine learning [47]. In environmental management, a large amount of information such as complex design and material information of buildings are related to environmental performance [48]. However, data in multiple formats poses difficulties in information management of construction projects [49].

As an information management method and technology oriented to the entire life cycle of construction project assets, BIM has also shown its advantages in environmental management [50,51]. The building's solar shadow, solar radiation, thermal comfort, indoor and outdoor ventilation, natural lighting, and air conditioning energy consumption can all be obtained through the analysis of the project BIM model [7,52]. This provides the basis for creating high-quality green buildings that are comfortable, healthy, energy-saving, waste-reducing, intelligent, and high-performance. In past research, BIM-based sustainability assessment approaches were developed [53]. More detailed aspects of environmental management of green or sustainable buildings, such as energy management [54], operational strategy [55], indoor air quality management [56] and daylighting analysis methods [57] are also developed. These studies demonstrate the potential of BIM technology in digital management and environmental assessment in construction projects. BIM's interoperability accommodates different disciplines and stakeholders, which facilitates collaboration and communication. It also enables cross-departmental cooperation in environmental management to be carried out effectively and quickly.

In ESG-related management, material selection and life cycle assessment, and waste management can all be handled through BIM models [19]. BIM can incorporate a database of sustainable materials and their life cycle impacts. This can provide data support for carbon emissions and carbon footprint management of construction projects [58]. At the same time BIM helps construction waste management by optimizing material usage, minimizing over-ordering and reducing on-site waste [5]. This is consistent with ESG objectives of responsible resource utilization [11]. However, how to accurately obtain the environmental information required for ESG assessment still requires further exploration. Accurate and rapid acquisition of relevant data can enhance the credibility of ESG reports and performance [33]. Current environmental-related research based on BIM is mainly focused on the analysis of the impact on construction projects and the optimization of related indicators [59]. In the processing of "E"-aware data (Environmental-aware data in ESG), although BIM can serve as an integrated platform to assist management, it still lacks domain knowledge for ESG in construction, which leads to the lack of comprehensive information statistics. For example, how to automatically capture carbon emissions, carbon footprint and other related content of construction projects for ESG assessment still needs further research. At the same time, processing this complex information is also one of the difficulties. In addition to the digitized environmental impacts of construction projects such as carbon emissions and pollution emissions, "E"-aware data also includes a large amount of text data related to the company's climate strategy, low carbon strategy and even government's environmental policy. This requires a large number of professionals and high manpower costs to process this information. LLM, as a natural language processing technology based on deep learning, allows the model to understand and generate text, helping practitioners to quickly process this information. It is a potential solution.

2.3. Potential of LLM for E-aware data processing

LLM, as a technology developed based on NLP, is a branch of it and has had a very important impact on society. Tasks including fluent language generation, language understanding and classification are more efficiently completed by LLM [60]. Since 2023, many LLMs have been developed and gained wide popularity. Models including ChatGPT, Llama and other models provide users with search engine, translation and other functions, demonstrating their powerful operating efficiency [61]. As the technical basis of LLM, NLP technology has been widely used in the processing of unstructured text in the past few decades [62]. During the development process, NLP-based text extraction technology was developed [63]. Behavioral assessment methods have also been upgraded by NLP technology [64,65]. Similar research provides the possibility for faster information management and analysis. However, traditional NLP is still difficult to solve the problems caused by the complexity and dynamics of language. Therefore, large-scale language models based on deep learning have been developed and provide easier processing of patterns. A large amount of high-quality text data is provided for model training [61]. So that it can further assist practitioners in their work. In the AEC industry, applications based on NLP or LLM have also been widely studied, as shown in Table 1.

These studies demonstrate the broad application prospects of LLMs or NLP models in the AEC industry. Similarly, in ESG management, related models also show advantages. Schimanski Tobias, et al. analyzed the gaps in corporate ESG measurement through NLP technology [77]. An NLP-based ESG consulting classifier has also been developed for corporate information management [78]. In addition, automatic ESG assessment methods using similar models have also been studied [79]. An approach to assessing a company's ESG-related behavior by analyzing comments on the Internet has also been devised [80]. These studies have provided extensive help for corporate ESG management.

Table 1
Recent Studies of NLP or LLM in AEC Industry.

No.	Models	Authors	Journals	Research focus
1	BERT	Zhu, X. et al. [66]	Automation in Construction	Knowledge mining
2	GPT	Xu, Qingfeng, et al. [67]	Advanced Engineering Informatics	Knowledge mining
3	GPT	Lin, W. Y. [68]	Sensors	Virtual assistant
4	Transformer	Zhang, R. et al. [69]	Automation in Construction	Knowledge mining
5	GPT	Zheng, J. et al. [20]	Automation in Construction	Virtual assistant
6	GPT	Prieto, S. A. et al. [70]	Buildings	Virtual assistant
7	BERT	Shuai, B. [71]	Computers in Industry	Match checking
8	Semantic Web	Pauwels, P. et al. [72]	Advanced Engineering Informatics	Match checking
9	BERT	Jung, Y. et al. [23]	Automation in Construction	Match checking
10	RNN and transformer	Shahinmoghadam, M. et al. [73]	Computers in Industry	Match checking
11	Improved TF-IDF algorithm in NLP	Wu, Z. et al. [74]	Expert Systems with Applications	Information retrieval
12	NLP trained by the CoSENT	Ma, Z. et al. [21]	Automation in Construction	Knowledge mining
13	GPT	Wan, Yuwei, et al. [75]	Advanced Engineering Informatics	Virtual assistant
14	GPT	Forth, K., & Borrmann, A. [76]	Journal of Building Engineering	Knowledge mining

However, there are still some difficulties in applying LLMs in the construction industry. In past studies, researchers mainly analyzed ESG reports that companies have released, or managed third-party information based on NLP models or LLMs. There is insufficient research on the processing of ESG-related information within enterprises. Existing related research cannot handle complex data sources in the AEC industry, such as engineering drawings and BIM models [81]. This information is an important management platform for construction projects, but it is difficult for non-professionals to directly process the relevant information. Therefore, the use of LLM in the AEC industry also faces two difficulties: (1) lack of methods to provide "E"-aware data of complex sources and formats to LLMs for processing, (2) lack of LLMs-based processing mechanism for complex structured data. Therefore, this paper presents an "E"-aware data processing framework based on BIM and LLM to support rapid processing of complex environment-related information in construction projects.

3. Methodology

The development of the "E"-aware data processing framework based on BIM and LLM follows the design science research methodology (DSRM), which is a problem solving paradigm that seeks to advance human knowledge by creating innovative artifacts [82]. Studies have adopted DSRM to develop procedures and information management solutions for the AEC industry [83,84]. Fig. 1 depicts a DSRM process to develop the "E"-aware data processing framework based on BIM and LLM for construction project ESG environment management with six steps:

(1) Step 1 presents the problem identification and motivation. The research problem is stated in Section 1 that there is a lack of an automatic "E" data processing approach using BIM and LLM for environmental management of ESG in AEC projects, resulting in complex and large amount data from construction projects.
(2) Step 2 presents the definition of the objectives. This work is aimed to develop automatic "E" data processing approach while fitting construction projects and ESG requirements. Three objectives are shown in Section 1, namely building an ontology-based knowledge model to specify "E"-aware data, developing an "E"-aware data sorting algorithm for BIM model, proposing an LLM-enhanced "E"-aware data processing mechanism.
(3) Step 3 presents the design and development. Section 3 displays the development process. Section 3.1 Based on the environmental indicators of the existing ESG assessment framework, summarizes the relationship between "E"-aware data in construction projects and ESG environmental indicators, and build ontology-based knowledge model for "E"-aware data for AEC projects. At the same time, the configuration of relevant data and the requirements for ESG assessment are analyzed to highlight the advantages and practicality of LLMs in its application. In Section 3.2, the "E"-aware data sorting algorithm for BIM based on visual programming is proposed, and the information derived from BIM in the established "E"-aware data requirements is extracted and organized to provide it to the final ESG data management and analysis. The developed algorithm has three functions: (i) "E"-aware data information supplement, adding carbon emission factor, construction material waste and other related content in the construction project to the BIM model for calculation, (ii) implementing BIM's "E"-aware data quantification, one-click quantification of information required for ESG such as embodied carbon emission in construction projects, (iii) organizing the output "E"-aware data, component location and construction based on BIM elements Time and other contents are used to express the relevant "E"-aware data in a structured manner to achieve convenient information traceability. According to the designed roadmap, Section 3.3 proposes an LLM-

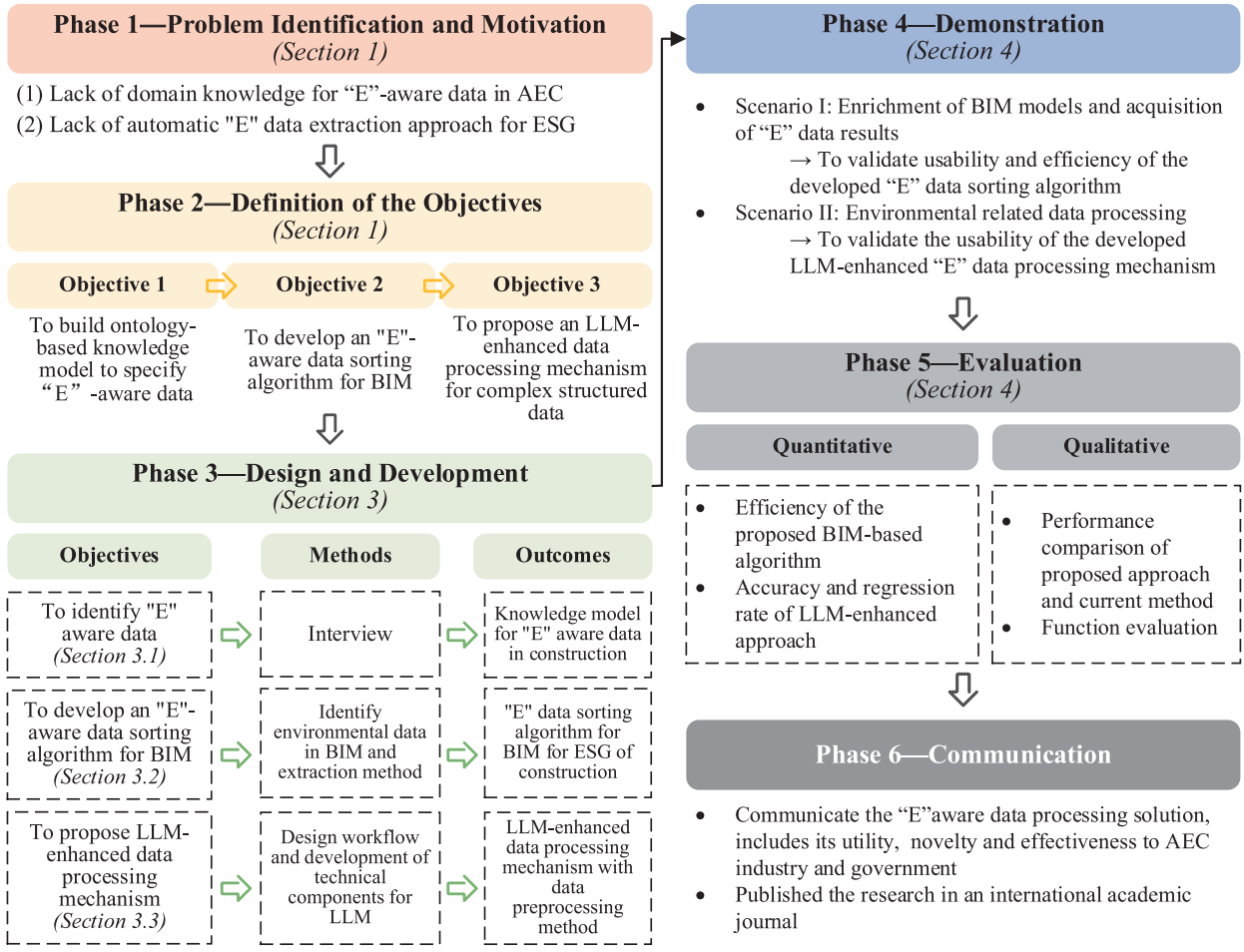


Fig. 1. Design Science Research Methodology (DSRM).

enhanced data processing mechanism for complex structured data. It implements three technical components: (i) construction data preprocessing kit. The sources of environment-related information in construction projects are complex and diverse. To support the operation of LLM including charts and construction drawings, this paper developed a data preprocessing solution. (ii) LLM-enhance mechanism technical component development. In the process of compiling ESG data, to ensure the accuracy of the report, the traceability of the data needs to be guaranteed. (iii) Integrated workflow. Due to the complexity of environment-related content management in construction projects, this paper proposes an "E"-aware data workflow. To achieve quick call, organization and use of environment-related information.

(4) Step 4 presents the demonstration. The proposed "E"-aware data processing solution in a design example is illustrated and its feasibility is verified (Section 4). The two proposed design scenarios validate the developed technical components.

(5) Step 5 presents the evaluation. Section 4 presents a quantitative and qualitative evaluation of the proposed "E"-aware data processing solution according to the requirements of DSRM to verify the usability of the proposed approach.

(6) Step 6 presents the communication. Researchers and industry professionals should understand the practicality and effectiveness of "E"-aware data processing solutions. and obtain relevant feedback and the development process is published in academic journals.

3.1. *Knowledge model for "E"-aware data in construction*

Before construction companies implement ESG, clarifying the generation process of ESG information will help to deal with relevant content in a targeted manner. In the daily operations of a company, various types of data will be generated, such as employee data, consumption data, purchasing data, etc., or non-quantified data, such as company systems, processes, etc. At this time, the company must collect this information. Sometimes even calculations are needed to obtain this useful ESG information. The disclosure of this information must be carried out in accordance with relevant standards to ensure that the public can obtain this content more effectively. The mainstream ESG disclosure frameworks currently on the market include Global Reporting Initiative (GRI), Corporate Sustainability Reporting Directive (CSRD), Task Force on Climate-related Financial Disclosures (TFCD), etc. These frameworks have their own requirements for different exchanges. At the same time, the ESG assessment framework helps investors and the public compare whether a company's ESG performance is good or not. These frameworks assign a score to a company's ESG performance by calculating and comparing its performance with other companies in the same industry. Through this process, investors can make sustainable investments and other financial behaviors based on these scores. Therefore, when some companies disclose their information according to the ESG disclosure framework, they will fail to disclose the content required in the assessment framework. The current mainstream ESG assessment frameworks include S&P, MSCI, FTSE, etc. The main environmental indicators of these ESG disclosure and assessment frameworks are shown in Fig. 2. The ESG frameworks in the figure have different focuses and objects [32]. For example, GRI is more oriented to stakeholders, SASB is

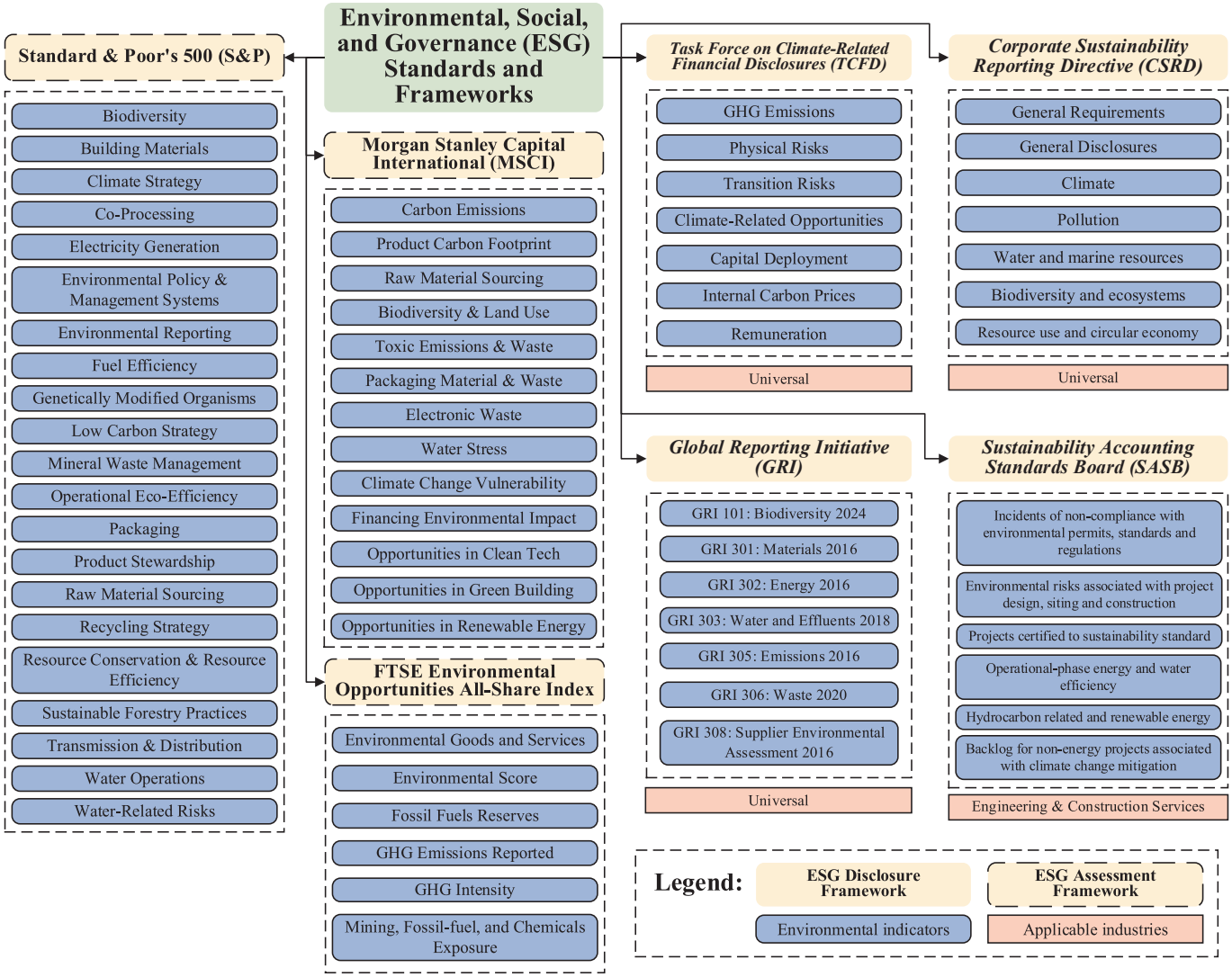


Fig. 2. Environmental indicators of Current ESG Frameworks and Standards.

oriented to investors, and TFCD will become a mandatory disclosure item on the Hong Kong Stock Exchange [85], which mainly considers climate change and its impact. These differences cause the information that companies disclose through different frameworks to differ from each other and make comparisons difficult [86]. As shown in Fig. 2, different ESG management or assessment frameworks emphasize different ESG data requirements. There are a large number of indicators that are the same or similar among them, but there are still some indicators that are only emphasized and required by some frameworks. Integrating these complex ESG management indicators and specializing them for the construction industry can help practitioners quickly manage and apply them to a wider range of frameworks as much as possible.

The concept of ESG is relatively new in the AEC industry, and most companies have been unable to develop appropriate methods to track ESG indicators in their construction projects, nor how to measure the ESG impact and effectiveness of projects [87]. At the same time, to face the complex environmental impact factors in the building construction stage, this paper is committed to developing an "E"-aware data indicator set that can track the environmental impact factors in the construction project. First of all, the design method is based on the consideration of possible environmental impacts during the building construction stage and the indicators of relevant institutions and researchers [88]. We follow the principles of scientificity, representativeness, and accessibility, and refer to existing ESG indicator design samples [89,90]. Existing environmental assessment indicators in ESG were screened out. Afterwards, a questionnaire survey was sent to 57 construction project managers or environmental managers from listed construction companies in mainland China and the Hong Kong Special Administrative Region. According to the characteristics of construction projects and the suggestions of professionals in the questionnaire, the existing environmental indicators in ESG management on the market were deleted, merged and modified to build a set of "E"-aware data indicators for construction projects. The questionnaire method and results are shown in Appendix A. At the same time, this paper refers to the initiative of the United Nations Sustainable Development Group, and the "E"-aware indicators developed are connected to 5 of the 17 goals related to the environment (Goal 7, 8, 11, 12 and 13) [91]. This provides clear assessment and management indicators for the qualitative SDGs targets in the sustainable development goals for the environmental impact of construction projects in the construction industry. Finally, ESG environmental assessment indicators and corresponding data sources based on the construction stage were designed, as shown in Fig. 3.

In this paper, the designed "E"-aware data indicators are divided into two categories, namely: (1) recourse and pollution, and (2) biodiversity and strategy. Prior to integration into our dataset, certain E-aware indicators necessitate computation and refinement. Part of recourse and pollution delineates the primary dataset required for construction

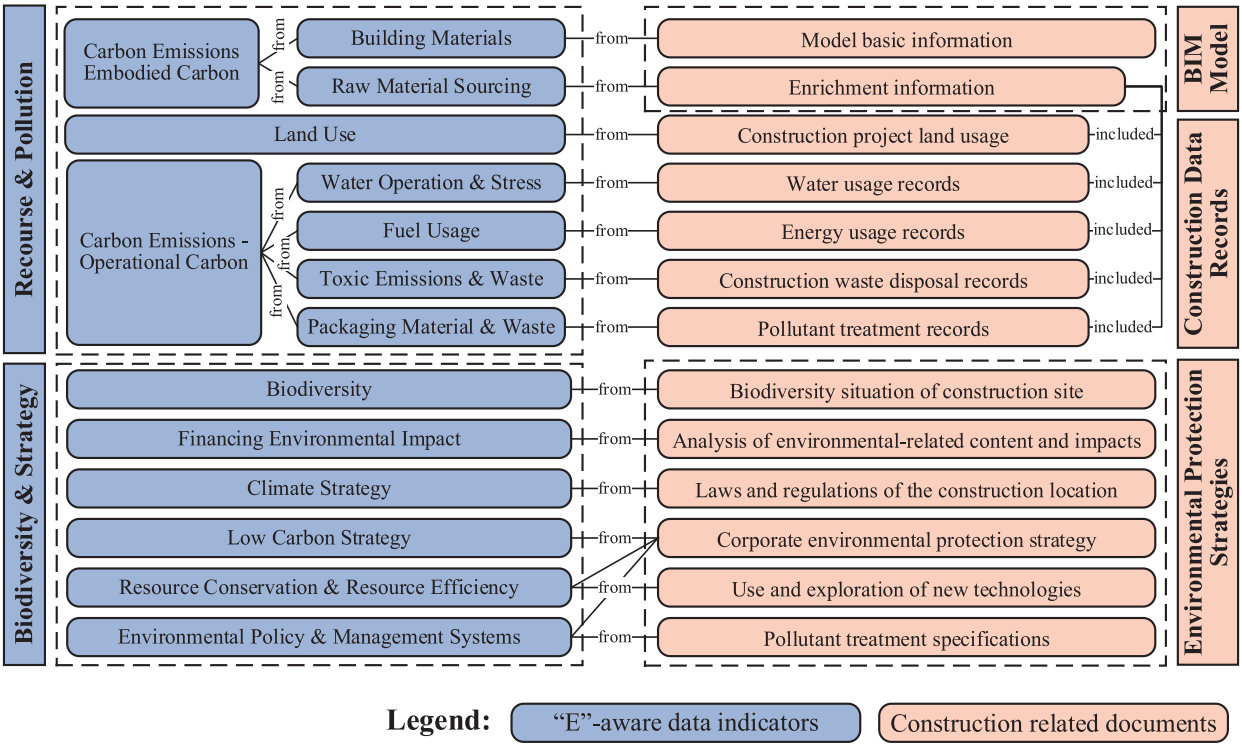


Fig. 3. Ontology of "E"-aware data indicators of ESG to construction data source.

projects, focusing on the records maintained during the construction phase. BIM, serving as the principal digital management tool, harbors extensive data conducive to environmental evaluation. Within these models, details such as material types and quantities are instrumental in supporting carbon emission assessments. Additionally, metrics on water consumption, waste generation, and energy usage derived from the construction activities can be incorporated into the BIM framework, leveraging actual project records. The BIM framework offers a construction element-based data architecture, enhancing the supplementation and traceability of "E"-aware data. Part of biodiversity and strategy emphasizes the need for "E"-aware data that encapsulates an organization's commitment to environmental stewardship. This encompasses strategic responses to climate change, adoption of low-carbon initiatives, and efficient resource utilization. Concurrently, the assessment of biodiversity effects at the construction site is also documented. Employing the ontology outlined herein, the interested parties can aggregate a comprehensive environmental profile for construction projects.

3.2. *"E"-aware data extraction algorithm for BIM*

To realize the sorting and extraction of "E"-aware data, this paper presents the development of an algorithmic tool, designed within the Dynamo visual programming environment, and seamlessly integrated into the BIM framework. Dynamo's efficacy in expediting and enhancing accuracy within BIM-centric research and environmental management practices is well-documented [92,93]. Advantages are also demonstrated in carbon emissions and related environmental management [94]. The tool developed in this paper can achieve two functions: (1) the supplement of environmentally related information to the BIM model, and (2) the calculation and extraction of "E"-aware data including carbon emission information. Fig. 4 illustrates the pseudocode for this algorithm. Initially, construction managers categorize construction records, align activities with corresponding BIM construction area elements, and enter this data into a database accessible to Dynamo. At the same time, information native to BIM and supplementary information (such as material types, supplier details, pollution indicators caused by construction activities, waste and energy consumption) are also integrated into specific indicators of the BIM model through the tool. Periodic input based on construction time can provide a basis for data quantification. BIM managers and designers can expediently enrich the model by selecting appropriate BIM family types or applying pre-defined parameters. The enriched BIM model underpins the second functionality of the tool: the extraction of E-aware data. Here, a pivotal component is the computation of carbon emissions, conducted per the modified methodology delineated in "Computation of Building Carbon Emissions," [95] which employs emission factor method. The revised calculation method is delineated as follows:

Carbon emission calculation formula for building materials [95]:

$$C_{JC} = \sum_{i=1}^{n} M_i F_i + \sum_{i=1}^{n} M_i F_s S_i \quad (1)$$

C_{JC}—Carbon emissions from the production and transportation stages of building materials (kg CO_2e).

M_i—Consumption of material i.

F_i—Carbon emission factor for building material type i (kg CO2e/unit of building material).

F_s—Carbon emission factor for the mode of transportation used for building material type i (kg CO_2e/(t•km)).

S_i—Transportation distance of building materials of type i (km).

Carbon emission formulae for the construction phase based on energy consumption [95]:

$$C_{JZ} = \sum_{i=1}^{n} E_{JZ,i} EF_i \quad (2)$$

C_{JZ}—Carbon emissions during the building construction phase ($kgCO_2$).

$E_{JZ,i}$—Total use of energy type i during the construction phase of the building (kWh or kg).

EF_i—Carbon emission factor for energy type i (kg CO_2/kWh or kg CO_2/kg).

Total energy use during the construction phase of the building

Step 1: "E"-aware data enrich to BIM model

Input: "E" enrichment information, BIM Model
Output: Enriched BIM Model
Step 1: Upload E enrichment information to Dynamo
Step 2: Detect "E" data type and kind of enriched information
if formatted as a "Carbon emission", "Pollution information" and Supply data
return *data ← Enrichment information*
if formatted as other information type
return *← Processing information*
else
return *distribution false*
Step 3: Enrichment information upload to BIM model
Enrichment information → Supplemented with fine-grained data of BIM components
Step 4: Filter the required Family type of BIM
Targeted Family Type ← Carbon emission factor||Material information||Supply chain data||Pollution information
Step 5: Processing information upload to BIM model
Processing information → BIM model overview
Step 6: Verify the integrity of uploaded information
if verification pass
output enriched BIM model
else
return *notification ← "E-aware information upload data is missing or wrong"*
End

Step 2: BIM-based "E" data quantification and extraction

Input: Enriched BIM Model
Output: "E" data set for construction
Step 1: Obtain basic project information in the BIM model
Step 2: Select the Family type in BIM model that needs to be collected
Step 3: BIM carbon emission and pollution calculation
Carbon emission data ← BIM component information × Carbon emission factor
Pollution data ← BIM material type × Pollution coefficient
Step 4: "E" data sorting of enriched BIM model
"E" data set ← "E" related data and information sources in BIM overview
Step 5: BIM information extraction
"E" data set ← BIM component information||Carbon emission data||Material type||Supply chain information||Pollution data
Step 6: Output the sorted "E" data set
End

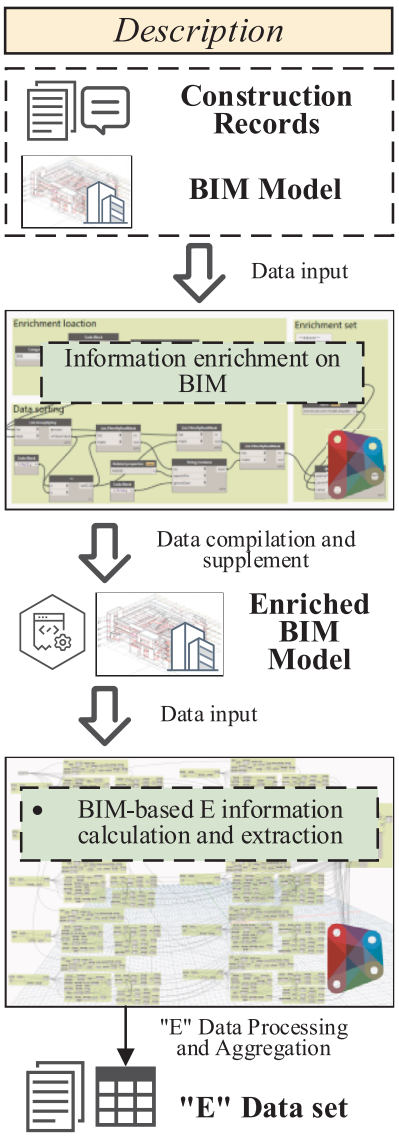


Fig. 4. BIM-based "E"-aware data extraction algorithm.

[95]:

$$E_{JZ} = \sum_{i=1}^{n} Q_{fx,i} f_{fx,i} + \sum_{i=1}^{n} Q_{cs,i} f_{cs,i} \quad (3)$$

$$f_{fx,i} = \sum_{j=1}^{m} R_j + E_{jj,i} \quad (4)$$

$$f_{cs,i} = \sum_{j=1}^{m} R_j \quad (5)$$

$$C_{JZ} = \sum_{i=1}^{n} Q_{JZ,i} f_{JZ,i} EF_i \quad (6)$$

E_{JZ}—Total energy use during the construction phase of the building (kWh or kg)

$Q_{fx,i}$—Quantity of work for item i of the sub-projects.

$f_{fx,i}$—Energy consumption factor for item i of the sub-project (kWh/unit of work).

$Q_{cs,i}$—Quantity of work for item i of the measures project.

$f_{cs,i}$—Energy consumption factor for item i of the measure project (kWh/unit of work).

R_j—Energy consumption of the jth construction machine (kWh).

$E_{jj,i}$—Total energy used by small construction equipment (kWh).

i –Project number

j –Machine serial number

Through the calculation of building materials, energy use during construction and carbon emissions corresponding to energy, the carbon emissions of construction projects (including embodied carbon and operational carbon) can be obtained. At the same time, other relevant "E"-aware data, including pollution information, supply chain data, etc., can also be extracted. This sorted information can be directly provided to the environmental management department or ESG personnel for information management and analysis.

3.3. LLM-enhanced "E"-aware data processing mechanism

In addition to formatted data such as BIM models, there is still a large amount of text and unformatted information such as engineering drawings that need to be processed in ESG management. This information often requires a lot of labor costs to sort out, which causes inconvenience to the company's ESG management. Therefore, based on the advantages of LLMs' fluent language understanding and generation, this paper proposes an LLM-enhanced data processing mechanism for complex structured data based on the modular system design principle [96], as shown in Fig. 5. Based on the proposed conceptual framework, a framework for "E"-aware data processing is further designed. The

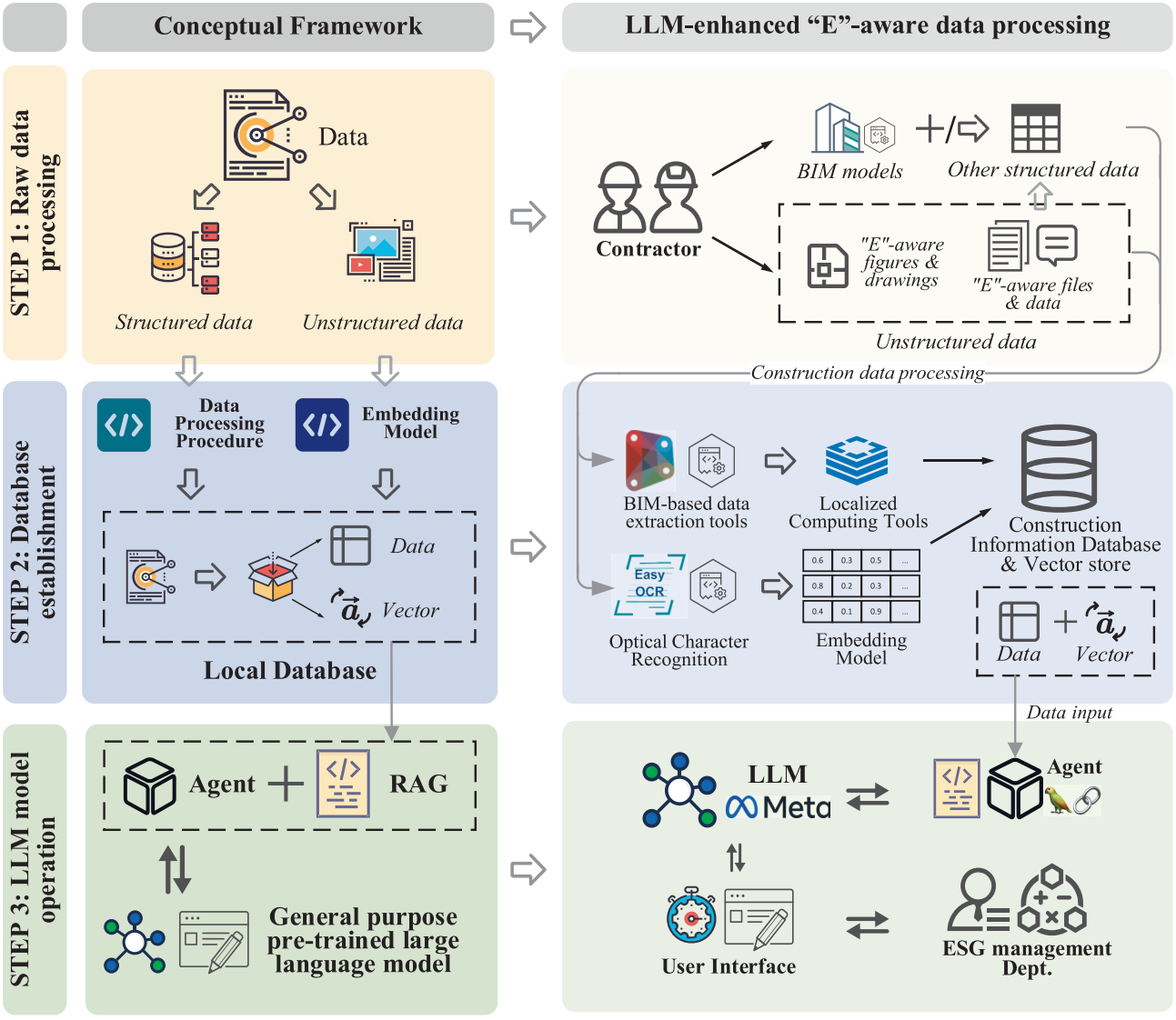


Fig. 5. LLM-enhanced data processing mechanism for complex structured data.

mechanism is divided into three steps. In step 1, the mechanism first collects all data sources so that they can be linked to the local database and provided to the LLM. These "E"-aware data are divided into three categories: information in the BIM model, environment-related construction drawings, and document information. Construction drawings include land use, construction plans, etc. These data are important sources of information in corporate ESG management. Obtaining similar information and confirming data sources are the basis for ensuring the integrity and authenticity of the report. In step 2, the local database is further organized through local computing tools. In this scenario, the information in the BIM model is obtained through Dynamo's BIM-based "E"-aware data extraction algorithm. The selected BIM platform is Autodesk Revit 2025 [97]. Data supplementation and structured acquisition will be carried out based on this platform, and the output Excel and XML format files will be stored in the local database. The unstructured data part will be preliminarily processed based on Easy-OCR. EasyOCR is a pre-trained text recognition model based on machine learning [98]. It supports more than 80 types of text and can accurately obtain text information, which is beneficial to LLMs' understanding and output of this information. In this part, OCR processing covers all types of engineering documents, and the recognized content is the text part of the drawings or designs in construction activities and the topography-related drawings. Only when the recognition confidence is > 0.85 is it included in the downstream LLM analysis to avoid noise transfer. At the same time, the processed unstructured text segments are converted into vectors through the embedding module and stored in the vector database.

Step 3, the local database is linked to the LLM. In order to meet the security requirements of construction data and ESG-related business data, the selection of LLM requires that it has the ability to be deployed locally, the ability to process information without networking, and the ability to retrieve from local databases. The LLM used in this paper is Meta's Llama 3 [99]. As an open-source large language model, it can achieve localized deployment, convenient fine-tuning instructions, and stable functions. In past studies, the Llama model has shown advantages in construction design [100], secure text processing [101], and environmental management [102], which is why it was chosen for this study. At the same time, the mechanism will use a Langchain-based agent to process local data and confirm data source information [103]. ESG management departments or other environmental managers will be able to obtain "E"-aware data conveniently and quickly through the interface, and can also check the source of the information to ensure the accuracy of the data processed by the LLM. When environmental managers query through the designed mechanism, the content stored in the vector database will be compared with the proposed themes and ESG-related indicators through cosine similarity test, and the ten most

relevant contents will be output. The cosine similarity test in this part ignores the differences in the modulus lengths of text segments and focuses on the consistency of semantic direction, [104,105] which is more suitable for comparing text segments of different lengths in construction projects. Through the development of the above steps, a virtual assistant based on "E"-aware data for construction is implemented. This mechanism uses locally deployed LLM and databases, which have certain professional information processing capabilities. At the same time, it does not need to upload enterprise and project data to the cloud, which can prevent users from unconsciously uploading enterprise information as LLM training data sets to the Internet, thereby ensuring that sensitive and confidential information of construction projects is not leaked.

To address the problem that it is difficult to ensure the accuracy of output information when using large language models, this paper

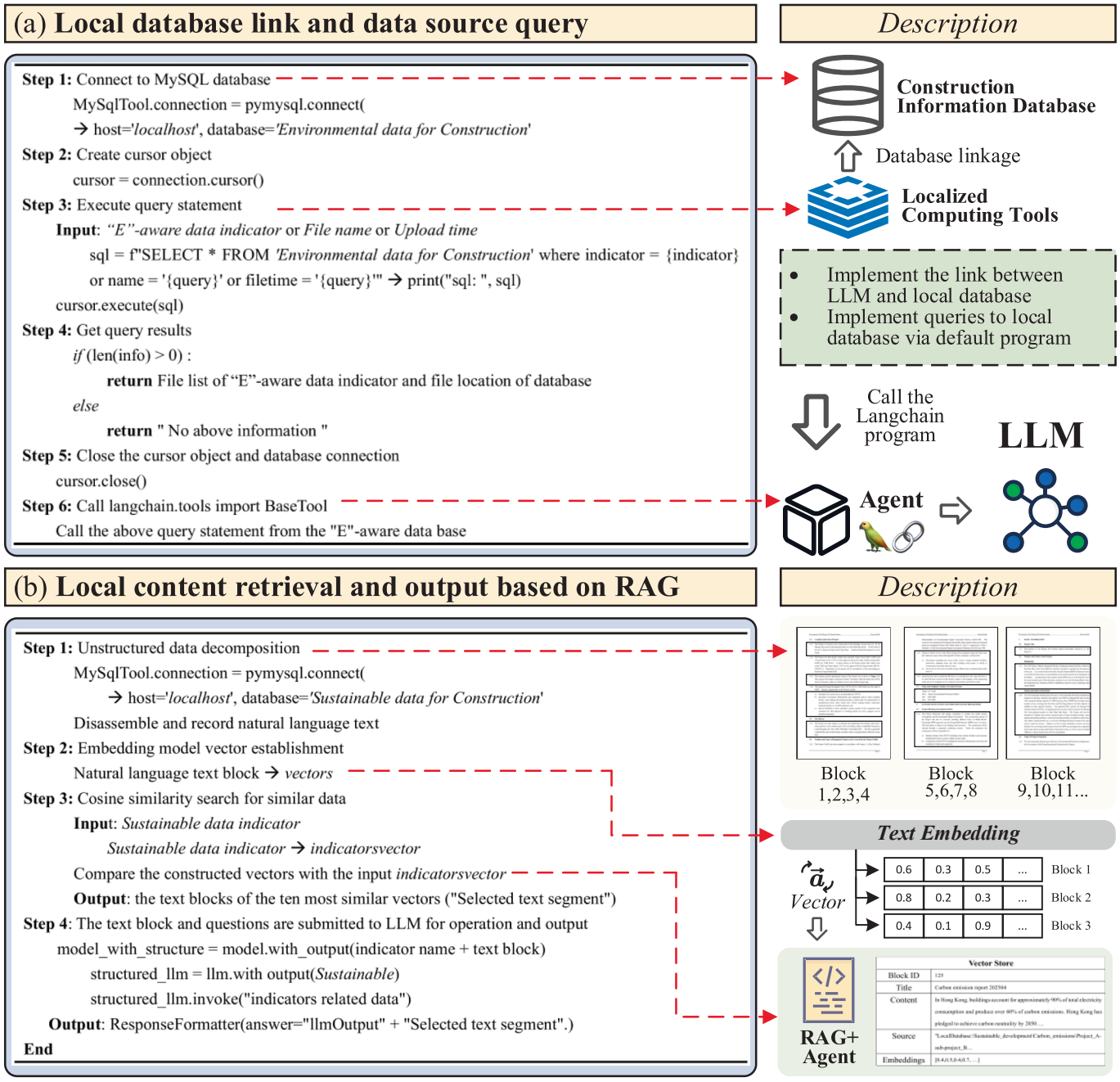


Fig. 6. (a) On-premise computing tool for LLM-enhanced data processing mechanism; (b) RAG-based unstructured data processing algorithm for LLM-enhanced data processing mechanism; (c) Data source query prompt words;

develops some technical components. It implements two functions: (1) LLM accesses and reads local database information and outputs the data source, as shown in Fig. 6 (a). Through Langchain's modular SQL database processing method, the program can process internal information locally through the design of read-only permissions, preventing LLM from reading confidential information without authorization and outputting unexpected content. The local calculation tool function is called through the function prompt. This mechanism enables environmental managers to confirm the source of information related to environmental data; (2) A RAG-based unstructured data processing algorithm is designed, as shown in Fig. 6 (b). RAG (Retrieval-augmented generation) allows LLM capabilities to be extended to a specific domain or organization's internal knowledge base without having to retrain the model [106]. This is a cost-effective way to improve LLM outputs so that they remain relevant, accurate, and useful in a variety of contexts [75]. By processing unstructured construction data, environmental managers can compare the required ESG indicators with the fields in the vector database and obtain more accurate results. This function enables managers to quickly obtain well-organized and readable "E" perception data with the help of LLM while ensuring the accuracy of information, meeting the readability and authenticity requirements of corporate ESG disclosures and reports.

4. Validation and evaluation

4.1. Validation method

This study applies the proposed automatic "E" data extraction approach to a design example to illustrate its feasibility and performance. The methodology of the proposed approach evaluation is shown in Fig. 7. For the validation process of the LLM-enhanced "E"-aware data management approach, the first step will be to validate the availability of "E"-aware data management in construction. In the validation of Section 4.2.1, the specific metrics include (1) "E"-aware data can be accessed, and (2) LLM-enhanced mechanism is available to the design scenario. In terms of performance evaluation, existing studies [20,70] usually measure the accuracy of data acquisition and LLM output. Therefore, to ensure the efficiency of the proposed method, in section 4.2.2, this paper evaluates (1) the accuracy of data collection and output, and (2) the efficiency comparison between the proposed method and traditional methods.

The testing environment is described as follows: (1) Windows 11 23H2 with Intel(R) Core (TM) i7-12700 K 3.61 GHz and 32 GB memory is used to develop large language model. (2) Llama 3 is selected as the large language model in the "E" data extraction approach. (3) Docker-based LLM and related program configuration, and (4) Revit 2021, Dynamo 2.1, and Python 2.7.0 within the developed dynamo program.

4.2. Qualitative and quantitative evaluation result

The verification scenario for the automatic "E" data extraction approach is based on the modified O·PARK2 (Organic Resources Recovery Center Phase 2) project from Hong Kong. The text data in the project comes from some construction documents and public environmental impact assessment reports. Fig. 8 shows the roadmap of verification scenario 1, whose purpose is to verify the usability of the "E"-aware data extraction algorithm in the developed solution. In addition to directly displaying raw data, environmental information needs to be calculated before it can be used in ESG disclosures. Among them, carbon emissions are representative because they are related to the data of many other indicators. First, contractors enrich relevant information into BIM (Step 1). The developed program then automatically calculates the carbon emissions generated during the construction project (Step 2). At the same time, other data based on BIM elements in "E"-aware data indicators can also be extracted and organized by the program.

The results in Fig. 8 show that the required supplementary environmental information is displayed in the BIM model, and the calculated and organized "E"-aware data is successfully generated. The calculation results of the carbon emission part are shown in Table 2. This paper compares the accuracy of calculation results. The emission factor used comes from the CIC carbon assessment tool database of Hong Kong [107]. Dynamo output is highly consistent with the manual calculation results (error < 0.03 %), verifying that the automated method meets professional accuracy standards. At the same time, automatic calculation has higher efficiency (save more than 90 % of manual calculation time), and the tracking of disclosed information can also quickly obtain supply chain information, construction waste and other related environmental content based on the structure of tabular data. When processing large amounts of construction data points, manual labor is prone to fatigue errors, while automation maintains consistency [108]. Automated calculations can maintain the traceability of workflow information [109]. The entire automation process leaves traces, avoiding random errors in manual operations during manual calculations (such as mis-entry of decimal points, formula overwriting, etc.).

Fig. 9 shows the roadmap of scenario 2, designed to verify the operability of the workflow based on LLM-enhanced "E"-aware data processing. In step 1, contractors upload environment-related documents and information to the database. Among them, land use, carbon emission information and other content obtained from the BIM model are processed in the step of scenario 1. Information appearing in Construction drawings is processed by EasyOCR to obtain the environmental data appearing in them. After all the information is uploaded to the

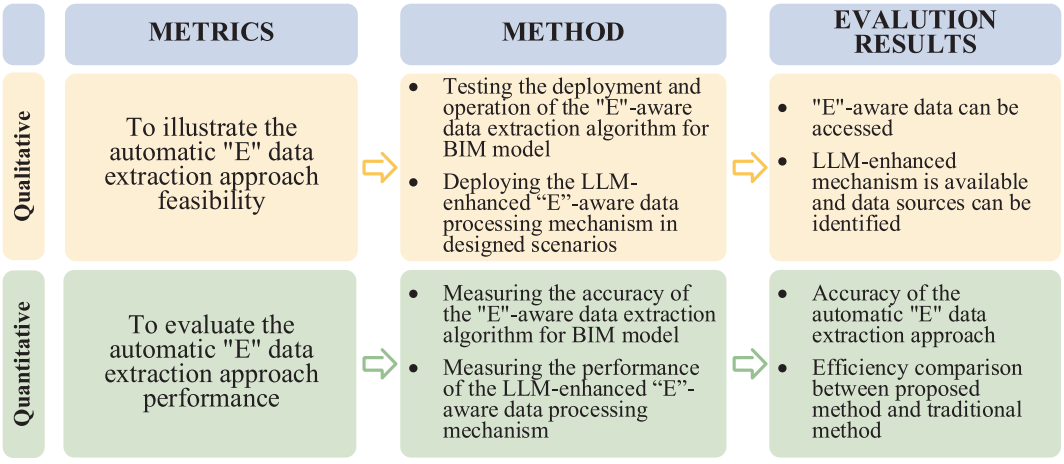


Fig. 7. Proposed approach validation and evaluation method.

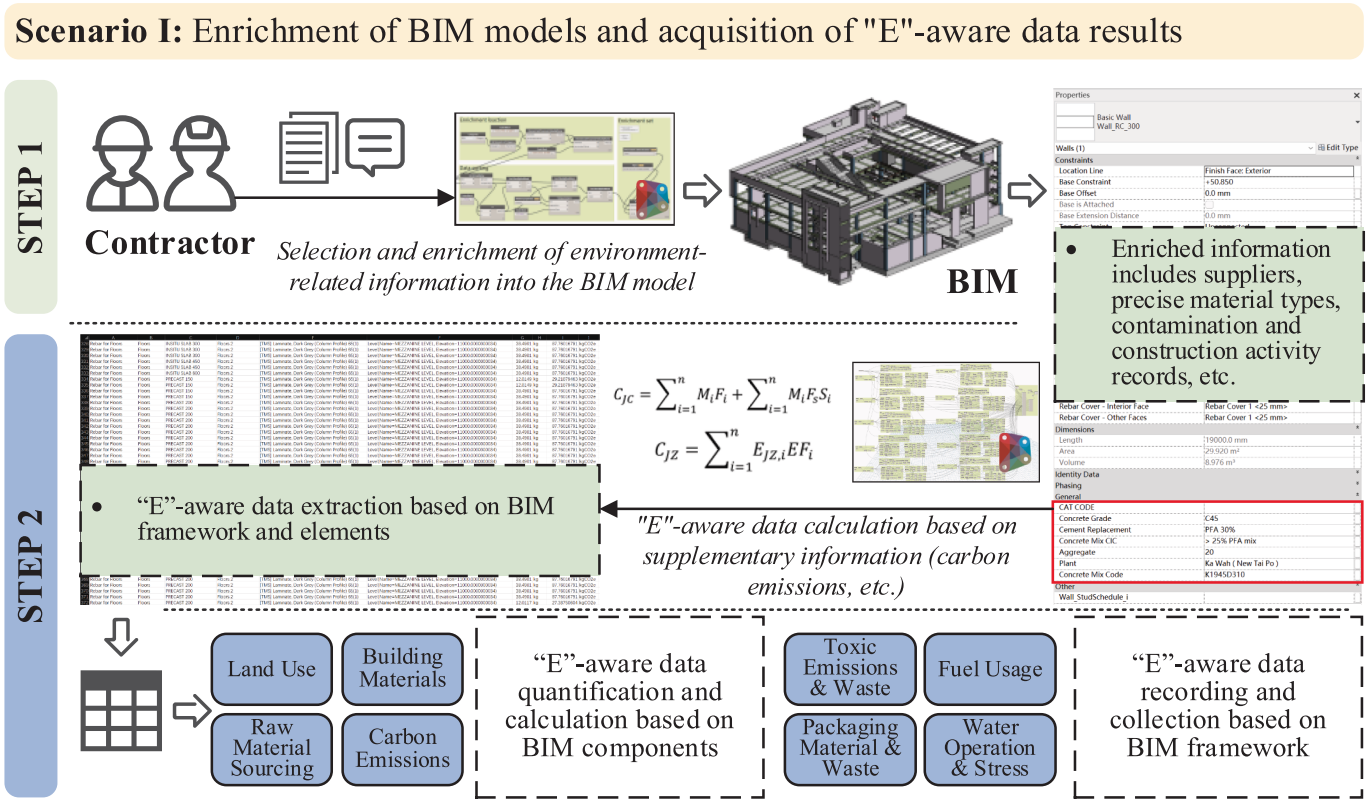


Fig. 8. Scenario I: Enrichment of BIM models and acquisition of "E"-aware data.

Table 2
Comparison between "E"-aware data extraction algorithm for BIM and manual calculation methods.

No.	Element	Material	Emission Factor	Manual calculation			"E"-aware data extraction algorithm		
			LCA: A1-A4	Quantity	Unit	Emission (t CO2-e)	Quantity	Unit	Emission (t CO2-e)
1	Foundation Slab	C45 30 % PFA	332.68 $kgCO_2/m^3$	7549.950	m^3	2,511.718	7549.967	m^3	2,511.723
2	Staircase	C45 30 % PFA	332.68 $kgCO_2/m^3$	186.920	m^3	62.198	187.254	m^3	62.296
3	Slab	C45 30 % PFA	332.68 $kgCO_2/m^3$	1843.120	m^3	613.299	1847.301	m^3	615.144
4	RC Column	C45 30 % PFA	332.68 $kgCO_2/m^3$	1187.500	m^3	395.059	1187.533	m^3	395.069
5	RC Wall	C45 30 % PFA	332.68 $kgCO_2/m^3$	3738.480	m^3	1,243.683	3737.629	m^3	1,243.434
6	RC Beam	C45 30 % PFA	332.68 $kgCO_2/m^3$	2710.910	m^3	901.743	2707.807	m^3	900.834
7	Mass Concrete Fill	C20 40 % GGBS	241.0875 $kgCO_2/m^3$	1751.630	m^3	422.171	1747.279	m^3	421.247
8	Formwork	Plywood	1.37 $kgCO_2/m^2$	72397.740	m^2	99.035	72397.740	m^2	99.035
9	Steel Beam	15 % Recycled Steel	2399.53 $kgCO_2$/ton	0.314	Ton	0.749	0.314	Ton	0.749
10	Steel Column	15 % Recycled Steel	2399.53 $kgCO_2$/ton	4.726	Ton	11.275	4.726	Ton	11.275
11	Rebar	15 % Recycled Steel	2399.53 $kgCO_2$/ton	3378.813	Ton	8,153.289	3378.933	Ton	8,153.568
12	LCA: A5 Construction Activities					904.014			904.014
Total						**15,318.270**			**15,317.804**

database, the on-premise computing tool developed in this paper connects it to the configured large language model (Step 2). In this step, the database provides accessibility to original documents to meet data traceability and information authenticity requirements for subsequent ESG disclosure requirements. In the configuration of LLM, this information is converted into TXT format and vector information and provided to LLM for model adjustment to suit the project's suitability. In step 3, managers of ESG departments or environmental-related matters (environmental managers) can obtain this information directly through LLM's user interface or command prompts. The LLM used in this designed scenario is Meta's Llama 3 [99]. In step 3 of Fig. 9, the left part shows the results of the user's inquiry about a specific indicator. In the case, waste information from the Opark2 project can be sorted and used directly in reports. The right part shows the results of confirming the indicator-related information source. The file name and file location of the original document are displayed in the results.

To gain a deeper understanding of the performance of the automatic "E" data extraction approach in environment-related management, this paper statistics the performance of large language models based on the accuracy of confusion matrix method, as shown in Table 3. The accuracy of "E"-aware data obtained based on LLM method and manual method will be based on this statistic. The "real" labels in confusion matrix created and reviewed by five senior domain experts provide a reliable and objective benchmark for evaluating the extraction accuracy of LLM tools and human participants. The labels used to evaluate the accuracy were strictly created by independent, experienced (more than 3 years) construction domain experts (not practitioners participating in the experiment). The experts independently annotated the original project documents (drawings, specifications, contracts, reports, etc.). The generated "real" dataset was internally reviewed to ensure consistency



Fig. 9. Scenario II: Environmental related data processing and LLM operation.

Table 3
Confusion matrix.

	Actual positive	Actual negative
Predicted positive	TP	FP
Predicted negative	FN	TN

and accuracy. In the evaluation, we selected 12 construction practitioners and undergraduate students to search for relevant chapters and sentences in 175 documents based on "E"-aware data indicators and timed them. The test participants need to be able to represent the main potential user groups of the tool in the future and the existing core user representatives in order to evaluate the performance of the LLM-driven automated data extraction tool in the actual application environment. Among them, the experience differences of undergraduates in specific fields are relatively small, which helps to reduce the confounding effect of individual experience differences on task completion efficiency (especially the learning curve) to a certain extent. In addition, AEC industry practitioners have rich domain expertise and practical experience, and have the deepest understanding of the needs, pain points and scenarios of actual data extraction. Their evaluation is crucial to judging the practicality, accuracy and efficiency improvement of the tool in actual work scenarios. The recruitment channel for testers comes from the course system within the department of university and industry partners who recommend construction project environmental managers with at least 3 years of work experience in related fields. At the same time, the same questions were asked for the developed approach. The prompt used refers to the research of Zamfirescu-Pereira, J. D., et al [110]. In terms of the selection of "E"-aware data indicators used in the test, six of them were selected for this test because the relevant content was unstructured text. These indicators are biodiversity, financing environmental impact, climate strategy, low carbon strategy, resource conservation & resource efficiency, and environmental policy & management systems. In the verification results, it took participants an average of 45 min to complete all the questions, and the fastest participant completed the "E"-aware data in 25 min, which is related to her professional experience. However, the developed approach only took 15 min, and the results were affected by the computer configuration. In terms of accuracy, because the specific sentences of the answers provided by LLM are not the same, we tested 10 times after deleting the cache of the model and obtained 60 results. The output accuracy calculation method and results obtained from the test are shown in Table 3 and Fig. 10.

$$Accurary = \frac{TP + TN}{TP + FN + FP + TN} \tag{7}$$

According to the results in Figs. 11 and 12, it can be found that compared with the manual data management method, the proposed "E"-aware data management solution shows a good efficiency improvement effect. In terms of accuracy, the result showed an accuracy of 84.43 % in responses of scenario. The LLM method maintained comparable accuracy to the manual method. In another test, the LLM mechanism connected to the local database improved the accuracy by 13.8 % compared to the general LLM. At the same time, in the test, the LLM-based "E"-aware data processing method saved 64.6 % of working time. Among inaccurate answers, the main fallacies in LLM are: (1) the data is not expressed, and (2) irrelevant indicator information appears. In fact, according to the process designed in this paper, it is necessary to check the information output by LLM. In this process, the occurrence of error messages can be avoided. Through the above designed scenarios test, the developed approach can assist practitioners in quickly and accurately processing information related to the construction environment. At the same time, we use the Quality of Experience (QoE) evaluation method to objectively evaluate the user experience quality of the LLM-enhanced data processing mechanism in the construction environment management process [111]. The test participants of the proposed framework were asked about their overall experience and satisfaction with the system. The evaluation covers dimensions such as system accuracy, response time, degree of fit, ease of use, and overall user satisfaction, in order to provide a reference for future optimization and improvement of the system. The invited QoE test participants were all employees of the environmental management department of a Hong Kong construction company that participated in the aforementioned usability test.

According to the feedback from 20 participants, the system showed

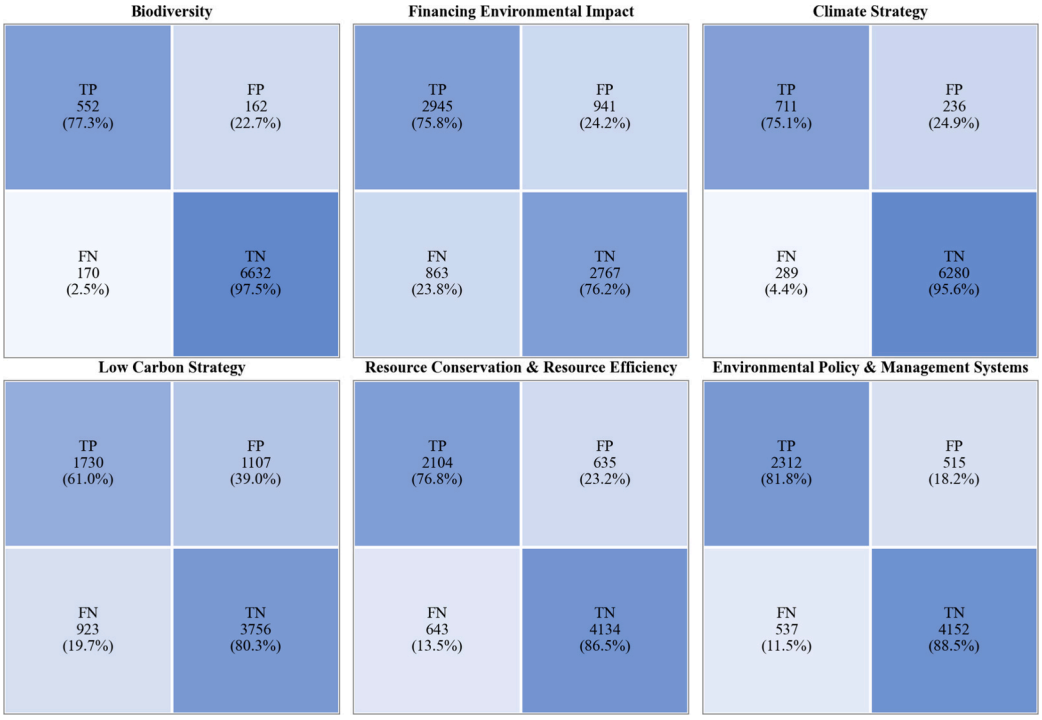


Fig. 10. Confusion matrix.

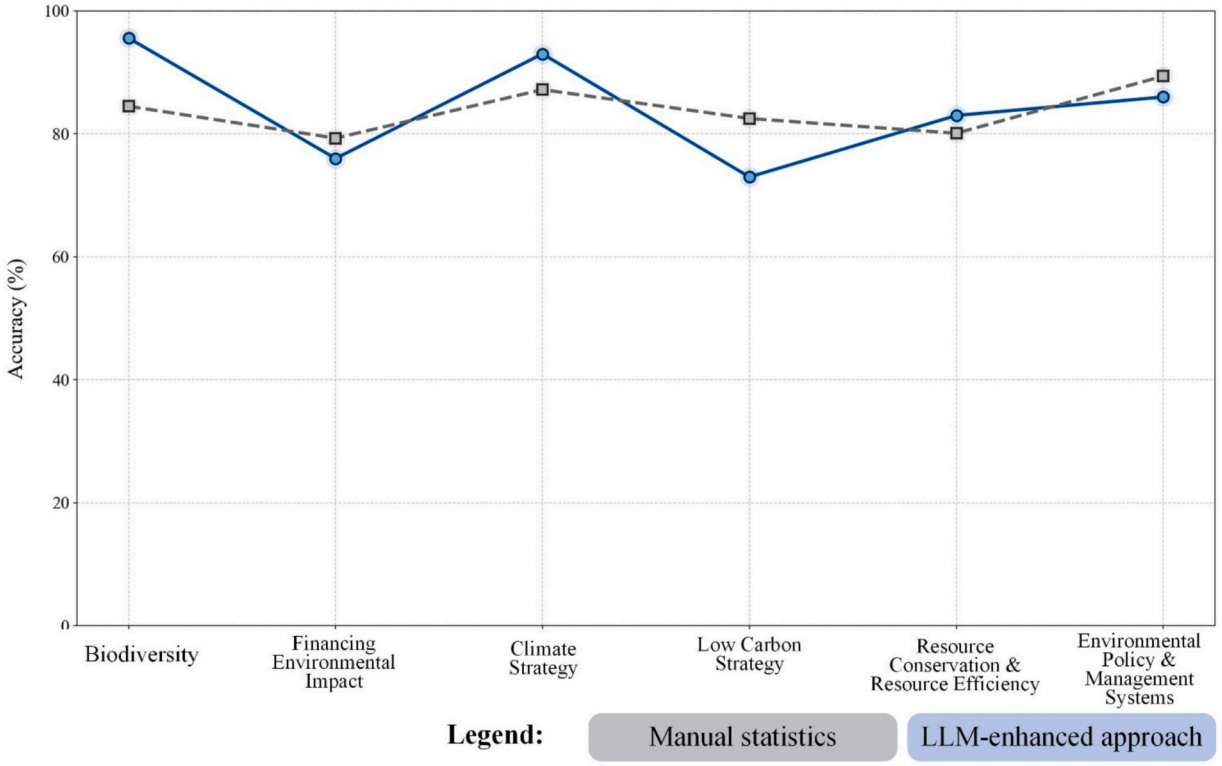


Fig. 11. Accuracy comparison between "E"-aware data manual-based approach and LLM-enhanced method.

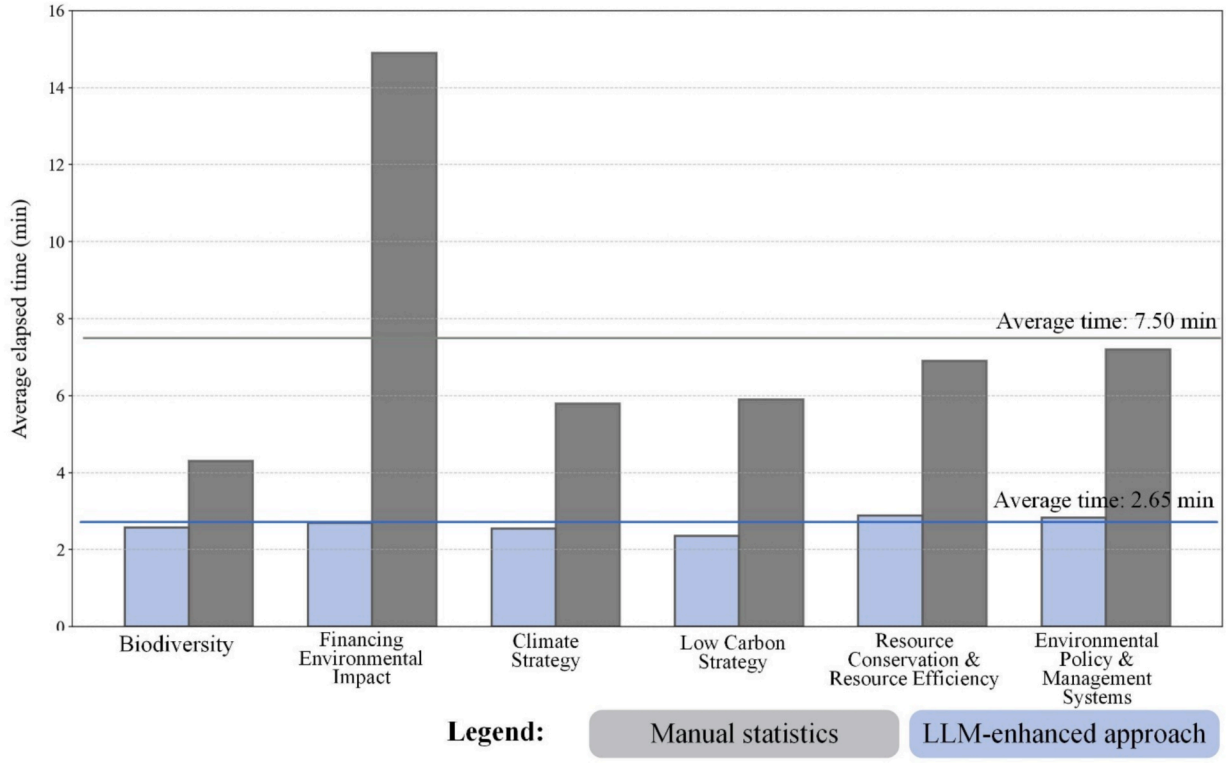


Fig. 12. Time comparison between "E"-aware data manual-based approach and LLM-enhanced method.

positive user satisfaction on several key performance indicators. The framework was evaluated on a five-point scale (5 for satisfied and 1 for dissatisfied), and the results are shown in Table 4. 90 % of the respondents were very satisfied or satisfied with the accuracy of the system, indicating that the system has a strong accuracy when handling operation and maintenance tasks. At the same time, some respondents also said that there is still room for improvement in the accurate representation of the output environment-related data. In terms of response

Table 4
Statistics on the results of the questionnaire.

	5	4	3	2	1
Accuracy	85 %	10 %	5 %	0 %	0 %
Response time	75 %	20 %	0 %	5 %	0 %
Degree of fit	80 %	20 %	0 %	0 %	0 %
Ease of use	75 %	15 %	10 %	0 %	0 %
Overall satisfaction	90 %	10 %	0 %	0 %	0 %

speed, the proposed framework has completed most of the data processing, making users relatively satisfied with this performance. At the same time, some respondents said that the response speed still needs to be improved. In terms of the applicability of the output information, the degree of fit also showed positive results. 90 % of the respondents expressed satisfaction with the ease of use of the proposed method. However, there were also suggestions that a better UI still needs to be developed. Overall, the proposed solution showed advantages in multiple indicators, and the improvements needed were also pointed out. Future solutions will further optimize these issues.

5. Discussion

5.1. Technical contributions

- **Developing an "E"-aware data extraction algorithm for BIM models.** BIM serves as a pivotal information management system in significant construction endeavors, offering essential data and structural support for the environmental assessment of AEC projects. This paper proposed a BIM-based "E"-aware data extraction technique that facilitates the swift sorting and retrieval of environmental information with a single click. This method enabled the augmentation, precise localization, and carbon emission computation directly from BIM components. Leveraging the BIM infrastructure, the algorithm exhibits flexibility, allowing for modifications to meet the diverse requirements of real-world engineering projects. It is adept at accommodating a multitude of scenarios, thereby enhancing its applicability. Furthermore, the algorithm contributes to the minimization of potential inaccuracies and oversights inherent in manual data compilation processes. By automating these tasks, it significantly improves the reliability and efficiency of "E"-aware data management in construction projects.
- **Providing an LLM-enhanced data processing mechanism for complex structured data**. It solves the challenge of handling complex situations where structured and unstructured data merge. The advent of LLMs has revolutionized data management processes, particularly in handling extensive textual content. This research synergizes LLM capabilities with ESG-related environmental information management, aiming to resolve the complexities and inefficiencies associated with traditional methods. Utilizing the advanced text comprehension, processing, and generation faculties of LLMs, the proposed system expedites the handling of large datasets. It introduces a consistent workflow that not only accelerates data processing but also ensures the integrity of "E"-aware data disclosure. To maintain the authenticity of the data and the accuracy of the output, we developed technical components to support the framework. These components facilitate the interrogation of local databases and verification of information sources, thereby reinforcing the authenticity of disclosed data. The implementation of this LLM-enhanced workflow promises a transformative impact on environmental management practices. It offers a robust solution to the erstwhile challenges of data processing, setting a new precedent for efficiency and reliability in environmental information management.

5.2. Theoretical contributions

- **Proposing a knowledge model for "E"-aware data in construction.** By scrutinizing current Environmental, Social, and Governance (ESG) disclosure frameworks and assessment criteria, this study formulates a set of 15 "E"-aware data indicators specific to the construction industry. The imperative for construction entities to mitigate their environmental footprint necessitates a robust mechanism for identifying and managing ESG-related aspects. This research is grounded in the tenets of scientific rigor, representativeness, and accessibility, ensuring the relevance and utility of the proposed indicators. The study delineates the derivation of these indicators, alongside validating their data sources. The indicators are designed to facilitate precise recognition of environmental concerns associated with construction activities within the AEC sector. The implementation of these indicators promises to furnish construction companies with a detailed framework for ESG environmental disclosure. It enables a granular approach to environmental management, aligning operational practices with sustainability objectives. The proposed knowledge model serves as a pivotal guide for construction companies striving to control the adverse environmental impacts of their operations. The clarity and specificity of the data maps empower companies to execute nuanced management strategies, thereby contributing to the broader goal of environmental stewardship.

5.3. Practical implications

- **Exploring the mechanism of integrating LLMs and construction data management.** The construction industry is increasingly turning to advanced technologies to streamline operations. LLMs, with their sophisticated data processing capabilities, offer promising solutions to enhance the efficiency and accuracy of construction project management. We propose a novel data management methodology that harnesses the power of LLMs, tailored to the specific needs of construction projects. The paper delves into the potential pitfalls associated with the deployment of LLMs in construction, proposing strategic solutions to mitigate these issues. It also outlines how the methodologies and design processes detailed herein can serve as a blueprint for developers seeking to create bespoke LLM-based applications in other domains. Through both qualitative and quantitative assessments, the paper demonstrates the benefits of the proposed LLM-integrated data management method. The findings **highlight** significant improvements in data handling processes within construction management. The integration of LLMs into construction data management presents a transformative opportunity for the industry. By leveraging the methodologies presented, developers can create more targeted and effective LLM-based applications, propelling the construction sector towards a more data-driven and sustainable future.
- **Considering methods for secure processing of construction management information based on large language models or AI.** With the further enhancement of digitalization in the AEC industry, the improvement of construction management by network-based AI models has also been noticed by researchers and practitioners. However, processing these construction-related information on public network platforms can lead to unexpected information leakage. In the use of LLMs, many companies and research institutes choose to avoid processing certain data directly through commercial LLMs. Because users do not know whether their data has been included in the training set of these models, other users can illegally obtain this information without knowing it. However, locally deployed large language models can avoid this situation. Especially in the complex construction management process, commercial secrets such as supplier information and financial statement information need to be treated more carefully. This paper implements a

method for internal processing of enterprise information through locally deployed LLMs and BIM-based algorithms, avoiding the risk of data leakage. Through the proposed solution, companies with similar needs can consider these methods to achieve more secure information management.

5.4. Limitations

- **Constraints in Dynamo-based Environmental Quantification.** The BIM-integrated "E"-aware data quantification algorithm relies critically on predefined material and activities libraries within Dynamo. While effective for projects utilizing standardized materials, this approach faces significant limitations when handling non-standard or innovative construction materials. The absence of pre-configured environmental impact coefficients (e.g., carbon emission factors, resource consumption metrics) for **unconventional** materials necessitates manual input – a process that is both technically demanding and potentially infeasible due to data scarcity. This dependency restricts the framework's adaptability to projects involving bespoke materials or emerging sustainable composites where authoritative lifecycle assessment (LCA) data may be unavailable. Further exploration of construction materials and activities in LCA management is necessary.
- **Multimodal data integration needs further exploration**. This study exclusively focuses on textual and structured environmental data processing, neglecting the integration of multimodal data (e.g., site imagery, sensor feeds, or material spectroscopy) – a critical capability of modern LLMs. Construction environmental management increasingly relies on heterogeneous data streams, where visual indicators (e.g., waste segregation compliance via site photos) or real-time IoT measurements (e.g., particulate matter sensors) provide essential "E" metrics. Future iterations should address this gap by developing cross-modal alignment mechanisms to synthesize visual, temporal, and textual evidence for comprehensive "E" assessment.

6. Conclusion

The management of "E"-aware data within construction projects is increasingly pivotal, particularly under the ESG paradigm. However, the processing of complex and large-scale ESG raw data in construction projects requires a lot of manpower and costs, which exacerbates the complexity of environmental assessment and disclosure in the AEC industry. LLM provides fast text understanding and processing capabilities and is a potential solution. Therefore, this paper proposed an automated "E"-aware data processing solution, leveraging LLMs to streamline environment-related information management within the AEC industry. Three objectives have been implemented. First, through screening and analysis of existing frameworks and standards, an ontology-based knowledge model was built to specify "E"-aware data, and a data map of environmental data sources and indicators was identified. Secondly, an "E"-aware data extraction algorithm for BIM model was developed. The proposed algorithm can supplement environment-related information according to the structural expression in the BIM framework and realize BIM-based "E"-aware data extraction and sorting. Finally, LLM-enhanced data processing mechanism for complex structured data using RAG is proposed. It has three key functions: (1) preprocessing of construction environment data for uniformity, (2) linking local databases with LLMs for comprehensive project information acquisition and analysis, and (3) developing an on-premise computing tool for data traceability and RAG-based data processing models to ensure the accuracy of ESG disclosure. This paper demonstrates the proposed automatic "E" data extraction approach for construction in a design scenario to verify its functionality. Both qualitative and quantitative assessments were conducted. The results show: (1) The extraction and processing of "E"-aware data based on BIM can be properly performed. The results are structured and displayed based on the BIM framework, which also provides the possibility for convenient data processing; (2) Compared with existing environmental management methods, the proposed approach enhances the efficiency of data management and saves time in processing construction projects time for a large amount of documentation; (3) the accuracy of the proposed solution is acceptable. These results demonstrate that the developed automatic "E" data extraction approach for construction and its technical components can support and enhance ESG-related environmental management of construction projects. The proposed method provides a feasible solution for processing multi-source and multi-format data in construction projects. In addition to environmental-related fields, this method can also be used as a reference in other possible complex processing scenarios.

As a preliminary exploration, the study acknowledges limitations due to current technological constraints. For instance, the potential for LLMs to provide erroneous information necessitates a robust verification process for "E"-aware data accuracy. To address these limitations and further refine the integration of LLMs with construction project management, future research directions include: (1) Improving the dynamic adaptability of the knowledge graph of "E" perception data to adapt to the update of information with indicators or construction and operation, and (2) Implementing targeted fine-tuning of LLMs to better adapt to the specific nuances and terminologies of the construction industry.

CRediT authorship contribution statement

Xingbo Gong: Writing – review & editing, Writing – original draft, Validation, Software, Methodology, Conceptualization. **Xingyu Tao:** Writing – review & editing, Validation, Methodology. **Yuqing Xu:** Methodology, Investigation. **Helen H.L. Kwok:** Writing – review & editing, Methodology. **Weiwei Chen:** Writing – review & editing, Investigation. **Da Shi:** Resources, Investigation. **Dezhi Li:** Writing – review & editing, Investigation. **Jack C.P. Cheng:** Supervision, Project administration, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A

Existing complex ESG environmental indicators struggle to accommodate the intricate data formats characteristic of construction projects. Consequently, seven widely recognized ESG management or disclosure frameworks were selected for further indicator screening. The screening

method adopted a questionnaire survey [112,113]. A total of 57 respondents (49 males and 8 females) were surveyed, all of whom were construction project managers or environmental management professionals in listed companies in mainland China and the Hong Kong Special Administrative Region.

Participants were asked: "Considering only the context of construction projects, which indicators do you consider relevant for assessing sustainability performance related to construction activities?" Each respondent was randomly assigned 2–4 current ESG management frameworks available on the market and tasked with evaluating each environmental indicator within their assigned frameworks as Relevant (Y), Not Relevant (N), or Undeterminable (N/A) for construction projects.

Responses with incomplete or contradictory information were excluded during data processing. The survey results pertaining to the seven ESG frameworks/disclosure schemes are presented in Tables A1–A7.

Table A1
Environmental Indicators and Construction Project Questionnaire Results for S&P.

Standard & Poor's 500 (S&P)		Y	N	N/A
1	Biodiversity	15	0	5
2	Building Materials	20	0	0
3	Climate Strategy	15	3	2
4	Co-Processing	2	3	15
5	Electricity Generation	0	4	16
6	Environmental Policy & Management Systems	17	0	3
7	Environmental Reporting	14	4	2
8	Fuel Efficiency	13	2	5
9	Genetically Modified Organisms	0	0	20
10	Low Carbon Strategy	20	0	0
11	Mineral Waste Management	3	0	17
12	Operational Eco-Efficiency	4	2	14
13	Packaging	16	2	2
14	Product Stewardship	14	0	6
15	Raw Material Sourcing	16	0	4
16	Recycling Strategy	15	1	4
17	Resource Conservation & Resource Efficiency	13	6	1
18	Sustainable Forestry Practices	2	0	18
19	Transmission & Distribution	0	2	18
20	Water Operations	15	0	5
21	Water-Related Risks	15	0	5

Table A2
Environmental Indicators and Construction Project Questionnaire Results for MSCI.

Morgan Stanley Capital International (MSCI)		Y	N	N/A
1	Carbon Emissions	20	0	0
2	Product Carbon Footprint	19	1	0
3	Raw Material Sourcing	18	0	2
4	Biodiversity & Land Use	16	0	4
5	Toxic Emissions & Waste	17	1	2
6	Packaging Material & Waste	17	2	1
7	Electronic Waste	0	11	9
8	Water Stress	16	2	2
9	Climate Change Vulnerability	15	3	2
10	Financing Environmental Impact	15	5	0
11	Opportunities in Clean Tech	19	0	1
12	Opportunities in Green Building	20	0	0
13	Opportunities in Renewable Energy	20	0	0

Table A3
Environmental Indicators and Construction Project Questionnaire Results for FTSE.

FTSE Environmental Opportunities All-Share Index		Y	N	N/A
1	Environmental Goods and Services	11	2	7
2	Environmental Score	9	0	11
3	Fossil Fuels Reserves	0	1	19
4	GHG Emissions Reported	9	1	10
5	GHG Intensity	15	0	5
6	Mining, Fossil-fuel, and Chemicals Exposure	1	0	19

Table A4
Environmental Indicators and Construction Project Questionnaire Results for TCFD.

Task Force on Climate-Related Financial Disclosures (TCFD)		Y	N	N/A
1	GHG Emissions	20	0	0
2	Physical Risks	9	4	7
3	Transition Risks	9	11	0
4	Climate-Related Opportunities	13	4	3
5	Capital Deployment	3	0	17
6	Internal Carbon Prices	2	0	18
7	Remuneration	0	0	20

Table A5
Environmental Indicators and Construction Project Questionnaire Results for GRI.

Global Reporting Initiative (GRI)		Y	N	N/A
1	GRI 101: Biodiversity 2024	16	2	2
2	GRI 301: Materials 2016	20	0	0
3	GRI 302: Energy 2016	20	0	0
4	GRI 303: Water and Effluents 2018	20	0	0
5	GRI 305: Emissions 2016	19	0	1
6	GRI 306: Waste 2020	20	0	0
7	GRI 308: Supplier Environmental Assessment 2016	17	0	3

Table A6
Environmental Indicators and Construction Project Questionnaire Results for CSRD.

Corporate Sustainability Reporting Directive (CSRD)		Y	N	N/A
1	General Requirements	5	13	2
2	General Disclosures	4	11	5
3	Climate	10	4	6
4	Pollution	19	0	1
5	Water and marine resources	9	10	1
6	Biodiversity and ecosystems	15	2	3
7	Resource use and circular economy	16	4	0

Table A7
Environmental Indicators and Construction Project Questionnaire Results for SASB.

Sustainability Accounting Standards Board (SASB)		Y	N	N/A
1	Incidents of non-compliance with environmental permits, standards and regulations	20	0	0
2	Environmental risks associated with project design, siting and construction	19	0	1
3	Projects certified to sustainability standard	19	0	1
4	Operational-phase energy and water efficiency	17	2	1
5	Hydrocarbon related and renewable energy	16	2	2
6	Backlog for non-energy projects associated with climate change mitigation	13	0	7

ESG environmental indicators achieving an endorsement rate exceeding 75 % in the survey were selected. These screened indicators, originating from various disclosure frameworks, were subsequently consolidated based on their conceptual similarity. The embodied carbon emission in the construction process comes from the calculation of the two indicators of building materials and raw material sourcing (LCA: A1-A4), and the operational carbon emission comes from the four indicators of water operation & stress, fuel usage, toxic emissions & waste, and packaging material & waste (LCA: A5). The final filtered set of "E"-aware data indicators relevant to construction is presented in Table A8.

Table A8
Filtered "E"-aware data indicators related to the construction process.

No.	"E"-aware data indicators for construction		MSCI	S&P	GRI	TCFD	FTSE	SASB	CSRD
1	Embodied Carbon	Building Materials	√	√	√	√	√		
2		Raw Material Sourcing	√	√	√	√	√		
3	Land Use		√					√	
4	Operational Carbon	Water Operation & Stress	√	√	√	√	√	√	
5		Fuel Usage	√	√	√	√	√	√	
6		Toxic Emissions & Waste	√		√	√	√		√
7		Packaging Material & Waste	√	√	√	√	√		√
8	Biodiversity		√	√	√				√
9	Financing Environmental Impact		√					√	
10	Climate Strategy		√	√				√	√
11	Low Carbon Strategy			√				√	
12	Resource Conservation & Resource Efficiency				√			√	√
13	Environmental Policy & Management Systems			√				√	√

Data availability

Data will be made available on request.

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